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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

AMCS101 – ALLIED MATHEMATICS - I

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. If $3 + i\sqrt{5}$ is a root of $2x^3 - 15x^2 + 46x - 42 = 0$ find the other roots.
2. State Cayley Hamilton theorem.
3. Prove that $cosh^2A - sinh^2A = 1$.
4. Find the nth derivative of e^{ax}.
5. Find the value of $\int x^4 e^x \, dx$.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Solve $x^3 - 12x^2 + 39x - 28 = 0$, given that its roots are in AP.
7. Show that the matrix $A = \frac{1}{3}\begin{pmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{pmatrix}$ is orthogonal.
8. Prove that $sinh^{-1}x = \log(x + \sqrt{x^2 + 1})$.
9. Find the radius of curvature at $\left(1, \frac{1}{2}\right)$ on the curve $2y = x(1 - x + x^2)$.

10. Find $\int_0^{\frac{\pi}{2}} \frac{sin^{3/2}x}{sin^{3/2}x+cos^{3/2}x}\,dx$.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) Solve $6x^5 - x^4 - 43x^3 + 43x^2 + x - 6 = 0$

(or)

b) Using Newton's method to find the real root of $x^3 + 3x - 1 = 0$ correct to 2 decimal places.

12. a) Investigate for what values of λ, μ the equations $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ have i) no solution ii) a unique solution (iii) an infinite number of solutions

(or)

b) Verify Cayley – Hamilton theorem for the matrix

$$A = \begin{pmatrix} 1 & -1 & 2 \\ -2 & 1 & 3 \\ 3 & 2 & -3 \end{pmatrix}$$

13. a) Prove that $cos8\theta = 128cos^8\theta - 256cos^6\theta + 160cos^4\theta - 32cos^2\theta + 1$

(or)

b) If $\sin(A + iB) = x + iy$ then show that

$$(i)\frac{x^2}{sin^2 A} - \frac{y^2}{cos^2 A} = 1\,,\,(ii)\frac{x^2}{cosh^2 B} + \frac{y^2}{sinh^2 B} = 1$$

14. a) If $y = \sin^{-1} x$, Prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2 y_n = 0$

(or)

b) If $x = rsin\theta cos\varphi$, $y = rsin\theta sin\phi$, $z = rcos\theta$ find the Jacobian of x, y, z w.r.to r, θ, φ

15. a) Find the reduction formula for $\int_0^{\frac{\pi}{2}} sin^n x\,dx$

(or)

b) Show that $\int_0^{\frac{\pi}{4}} \log(1 + tanx)\, dx = \frac{\pi}{8}\, log2$

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